Wealth Core Expectancy-Aware Persistent Bear Challenger

Functional Specification for Engineering Implementation

Research version: Iteration 2 | Specification date: August 6, 2026

Status: research challenger; not yet production-certified

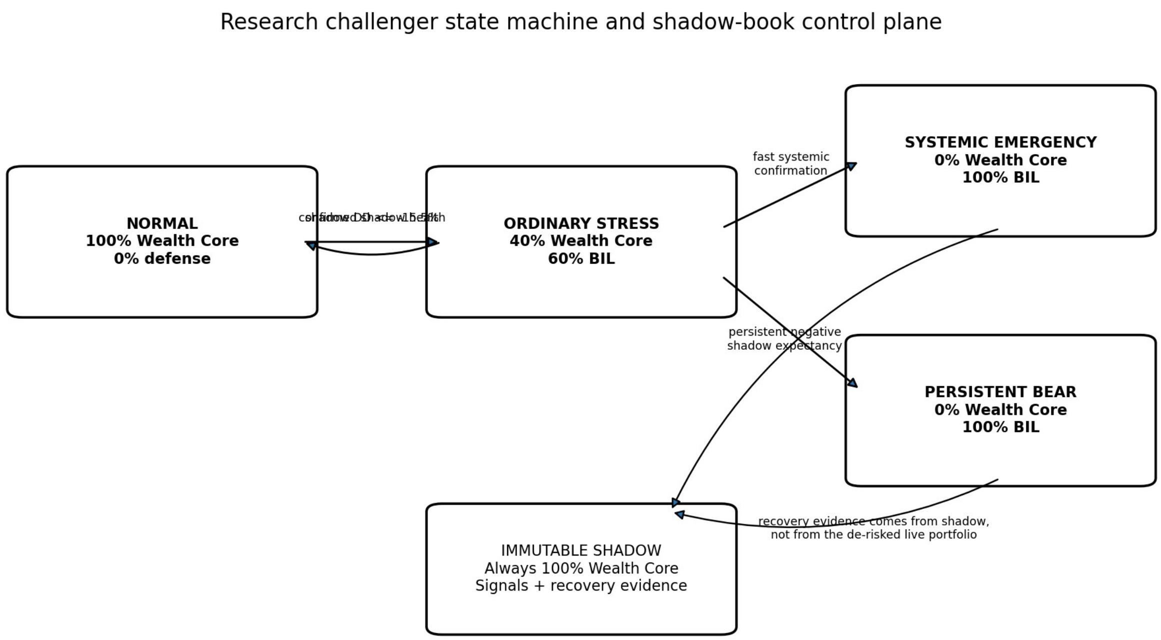

Figure 1. High-level state machine and immutable shadow control plane.

Purpose. This document defines the complete observable behavior, data requirements, state transitions, calculations, execution semantics, pseudocode, examples, logging, and validation criteria needed to implement the latest Wealth Core research challenger. It is written so an engineer can build the controller without relying on informal conversation history.

Important status. The strategy described here is a research artifact derived from frozen historical paths. It must not be represented as production-certified until exact next-open execution, data lineage, corporate actions, transition costs, state persistence, and parity with the trusted simulator have been validated.

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1. Executive summary

The latest challenger adds an expectancy-aware Persistent Bear state to the existing Wealth Core defensive architecture. The core strategy and its immutable shadow portfolio continue to run continuously. The live portfolio changes only its allocation between Wealth Core and a Treasury-bill defensive sleeve.

State	Wealth Core	Defensive sleeve	Primary purpose
NORMAL	100%	0%	Full participation in Wealth Core compounding
ORDINARY_STRESS	40%	60%	Reduce risk after a 15.5% shadow drawdown
SYSTEMIC_EMERGENCY	0%	100%	Exit during fast, externally confirmed market shocks
PERSISTENT_BEAR	0%	100%	Exit when the shadow remains persistently unproductive during prolonged stress

The central design principle is that all risk decisions and all recovery decisions use the immutable, fully invested shadow portfolio. The live portfolio is never allowed to judge its own recovery because its returns are altered by de-risking.

Selected research parameters:

Parameter	Value	Meaning
Minimum ordinary-stress duration	30 sessions	Persistent Bear cannot activate earlier
Shadow loss since stress began	<= -2%	Residual equity exposure has continued to lose
Shadow 40-session return	<= -3%	Medium-term expectancy remains negative
Positive-day share, last 20 sessions	<= 50%	No more than half of recent sessions are positive
Damaged holdings breadth	>= 75%	Broad internal impairment
Green holdings breadth	<= 25%	Few healthy holdings remain

Minimum Persistent Bear hold	20 sessions	Prevents immediate whipsaw
Recovery confirmation	5 consecutive healthy sessions	Requires sustained evidence before exit

2. Scope and design principles

2.1 In scope

- Allocation controller layered above Wealth Core.
- Immutable full-exposure shadow portfolio.
- Ordinary 15.5% portfolio backstop.
- Systemic-emergency zero-exposure override.
- Expectancy-aware Persistent Bear zero-exposure state.
- BIL or equivalent Treasury-bill total-return sleeve.
- Transition costs, state persistence, telemetry, and deterministic replay.

2.2 Out of scope

- Changing Wealth Core security selection, ranking, position sizing, or individual 30% stop logic.
- Using the de-risked live portfolio to calculate signals.
- Discretionary human overrides not represented as explicit, auditable events.
- Leverage, shorting, inverse ETFs, derivatives, or market timing based on news sentiment.

2.3 Non-negotiable principles

- Shadow continuity: the shadow remains 100% Wealth Core through every defensive episode.
- No look-ahead: all features at decision date t use information available by the decision timestamp.
- Single state owner: one controller instance owns state transitions for one portfolio.
- Determinism: identical inputs and configuration must produce identical transitions, orders, and NAV.
- Hysteresis: entry and recovery conditions are intentionally asymmetric.
- Explainability: every transition must carry a machine-readable reason and feature snapshot.

3. System architecture

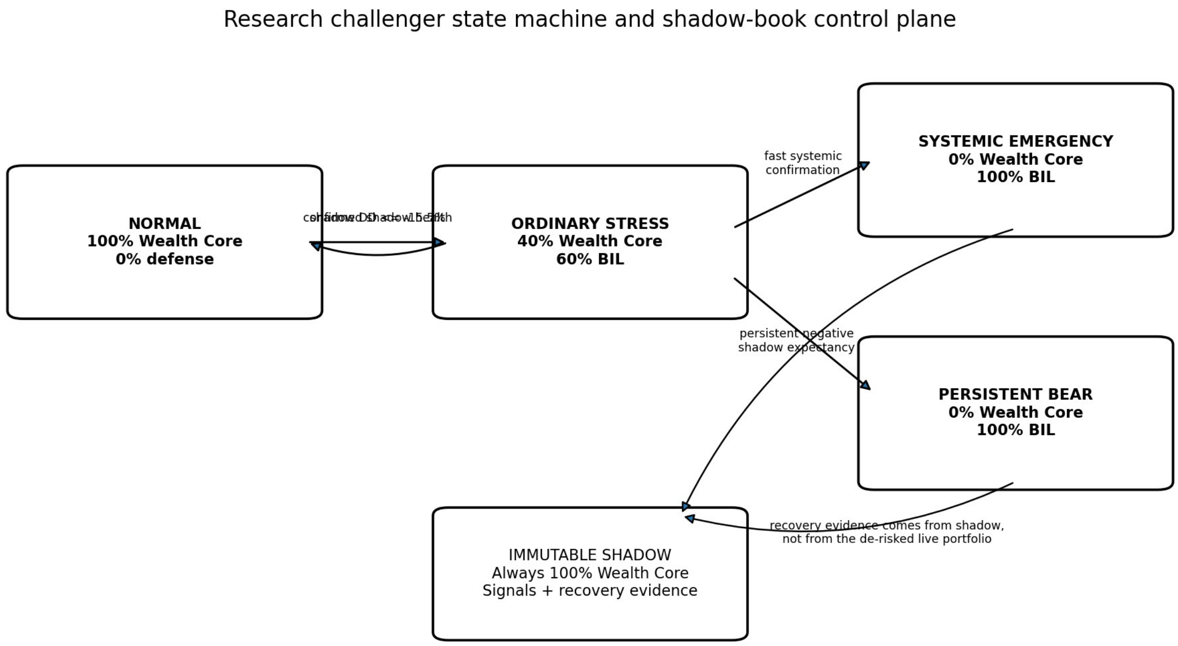

Figure 2. Functional architecture: live allocation is governed by conditions measured on the immutable shadow.

Component	Responsibility
Wealth Core engine	Produces daily target holdings and full-exposure shadow returns.
Shadow book	Tracks the hypothetical continuously invested Wealth Core portfolio and all holding-health metrics.
Feature engine	Computes drawdowns, multi-horizon returns, breadth, volatility acceleration, and positive-day share.
Ordinary controller	Moves live allocation from 100% to 40% when the shadow drawdown backstop triggers.
Systemic classifier	Detects fast internal shock plus external market confirmation; can force allocation to 0%.
Persistent Bear classifier	Detects prolonged negative shadow expectancy while ordinary stress remains active; can force allocation to 0%.
Allocation arbiter	Resolves competing states by priority and outputs one target Wealth Core allocation.
Execution adapter	Translates allocation changes into sell/buy orders for Wealth Core holdings and BIL.
Ledger and telemetry	Persists states, feature snapshots, orders, fills, costs, NAV, and audit events.

4. Data contracts and derived features

4.1 Required daily inputs

Field	Type	Definition
session_date	date	Canonical exchange session date.
shadow_nav	decimal	Total-return NAV of the immutable 100% Wealth Core shadow.

shadow_holdings	collection	Security IDs, weights, prices, peaks, entry state, and health classifications.
spy_adjusted_close	decimal	Adjusted total-return proxy used by systemic confirmation.
defense_return	decimal	Daily total return of the defensive sleeve.
ordinary_stress_flag	boolean	Whether the normal 15.5% backstop is active.
systemic_emergency_flag	boolean	Whether the systemic emergency state is active after hysteresis.
live_nav	decimal	Actual simulated or live portfolio NAV before the next rebalance.
current_allocation	decimal	Current live Wealth Core fraction in [0,1].

4.2 Derived features

Feature	Formula / rule
shadow_peak_t	max(shadow_nav_0 ... shadow_nav_t)
shadow_dd_t	shadow_nav_t / shadow_peak_t - 1
r5, r10, r20, r40	shadow_nav_t / shadow_nav_(t-n) - 1
positive_day_share20	count(shadow_return > 0 over last 20 sessions) / 20
damaged_breadth	damaged shadow holdings / valid shadow holdings
green_breadth	green shadow holdings / valid shadow holdings
damaged_breadth_delta5	damaged_breadth_t - damaged_breadth_(t-5)
SPY vol acceleration	annualized_std_5(SPY returns) / annualized_std_20(SPY returns) - 1
SPY r20	SPY adjusted close_t / SPY adjusted close_(t-20) - 1
stress_duration	consecutive sessions ordinary_stress_flag has remained true
stress_reference_return	shadow_nav_t / shadow_nav_at_stress_start - 1

Breadth classification contract. The controller consumes damaged and green labels from the certified Wealth Core shadow implementation. Their security-level definitions must be versioned and identical in research, rehearsal, and live operation. The controller must not silently substitute a different definition.

5. State machine and priority

5.1 State priority

When more than one condition is true, the allocation arbiter applies the most defensive state. Recommended priority: SYSTEMIC_EMERGENCY > PERSISTENT_BEAR > ORDINARY_STRESS > NORMAL. Both zero-exposure states produce the same allocation but retain distinct state identifiers and transition reasons.

Condition	Target allocation
systemic_emergency_active = true	0.00
else persistent_bear_active = true	0.00
else ordinary_stress_active = true	0.40
else	1.00

5.2 Ordinary stress

Entry. Activate when immutable shadow drawdown is at or below -15.5%. The ordinary controller is independent of the two zero-exposure overlays.

Action. Set live Wealth Core target to 40% and defensive sleeve to 60%. Continue running the shadow at 100%.

Recovery. Use the certified ordinary-controller recovery rule. In the research lineage this is a shadow-health rule with confirmation and hysteresis. The implementation must import the exact approved rule as a versioned dependency rather than re-infer it from this document.

5.3 Systemic emergency

The latest challenger changes the established systemic action from a 40% floor to a 0% floor. The systemic classifier itself uses the following research rule.

- Shadow drawdown <= -10%.
- Damaged breadth >= 85%.
- Green breadth <= 20%.
- Either shadow r5 <= -5% or shadow r10 <= -8%.
- Damaged breadth increased by at least 40 percentage points over five sessions.
- SPY 5-session realized volatility is at least 4% above SPY 20-session realized volatility.
- Either SPY r20 <= -1% or shadow r10 <= -10%.

Entry gating. The classifier must be armed, must not already be active, and in the historical implementation did not open a redundant emergency while the normal backstop already owned the same transition. For a production implementation, duplicate-state semantics must be explicit and tested.

Recovery. Minimum hold 10 sessions, then require three consecutive healthy sessions where shadow r20 > 0, damaged breadth <= 60%, and green breadth >= 20%. Re-arm only after the shadow drawdown recovers above -6% and no systemic signal is present.

5.4 Expectancy-aware Persistent Bear

Entry is evaluated only while ordinary stress is continuously active and Persistent Bear is inactive. All conditions must be true on the same decision session:

Predicate	Selected threshold
stress_duration >= min_stress_sessions	30 sessions
stress_reference_return <= loss_entry	-2%
shadow_r40 <= r40_max	-3%
positive_day_share20 <= positive_share_max	50%
damaged_breadth >= damaged_min	75%
green_breadth <= green_max	25%

Action. Set persistent_bear_active = true and target Wealth Core allocation = 0%. Record the stress-start index, entry index, all features, and rule version.

Recovery. After at least 20 sessions in Persistent Bear, define a healthy session as shadow r20 > 0, damaged breadth <= 60%, and green breadth >= 20%. Exit only after five consecutive healthy sessions. On exit, return to the allocation currently required by the underlying controller: 40% if ordinary stress remains active, otherwise 100%. Do not automatically jump to 100%.

Stress-reference handling. If ordinary stress becomes false, clear the stress-start reference. If Persistent Bear exits while ordinary stress remains true, set the stress-start reference to the exit session. This prevents immediate re-entry using stale pre-recovery losses.

6. Execution and accounting semantics

6.1 Decision timing

Research path semantics were close-to-close: features were observed on session t, allocation a_t was selected, and a_t governed the next return interval. A certified implementation should use explicit next-open orders unless the broader simulator standard specifies otherwise. The signal timestamp and fill timestamp must never be conflated.

6.2 Rebalance calculation

```
target_core_value = total_portfolio_value * target_allocation
core_trade_value   = target_core_value - current_core_value
defense_trade_value = -core_trade_value

# Sell one sleeve and buy the other.
turnover = abs(core_trade_value)
transition_cost = 2 * turnover * cost_per_side
```

Research experiments charged 10 basis points per side (cost_per_side = 0.001), therefore a complete sleeve switch incurs costs on both the sold and purchased sleeves.

6.3 Defensive asset

Use a short-duration government-bill total-return proxy. The research data lineage used cash before SHY availability, SHY until BIL availability, and BIL thereafter. A production implementation should preferably use BIL or a directly investable Treasury-bill instrument and must model dividends, adjusted prices, spreads, and settlement.

6.4 Position-level behavior

When live Wealth Core allocation is reduced, the live book should scale the complete target portfolio proportionally unless the execution design explicitly specifies a deterministic liquidation order. The immutable shadow continues to hold and manage the full Wealth Core portfolio, including entries, exits, reviews, individual stops, and corporate actions.

7. Reference pseudocode

```
CONFIG:
    ordinary_trigger_dd = -0.155
    persistent.min_stress_sessions = 30
    persistent.loss_entry = -0.02
    persistent.r40_max = -0.03
    persistent.positive_share20_max = 0.50
    persistent.damaged_min = 0.75
    persistent.green_max = 0.25
    persistent.min_hold_sessions = 20
    persistent.recovery_confirmation = 5

STATE:
    ordinary_stress_active: bool
    systemic_emergency_active: bool
    persistent_bear_active: bool
    ordinary_stress_start_session: optional session_id
    persistent_entry_session: optional session_id
    persistent_healthy_streak: int
    systemic_entry_session: optional session_id
    systemic_healthy_streak: int
    systemic_armed: bool
    previous_target_allocation: float

FOR EACH COMPLETED SESSION t:
    shadow.update(t)
```

```
f = feature_engine.compute_from_shadow_and_market(t)

# Ordinary stress reference tracking
if ordinary_stress_active(t):
    if ordinary_stress_start_session is None:
        ordinary_stress_start_session = t
else:
    ordinary_stress_start_session = None

# Persistent Bear entry
if not persistent_bear_active and ordinary_stress_start_session is not None:
    stress_days = sessions_between(ordinary_stress_start_session, t) + 1
    stress_loss = shadow_nav[t] / shadow_nav[ordinary_stress_start_session] - 1
    enter_persistent = all([
        stress_days >= 30,
        stress_loss <= -0.02,
        is_finite(f.r40) and f.r40 <= -0.03,
        is_finite(f.positive_day_share20) and f.positive_day_share20 <= 0.50,
        f.damaged_breadth >= 0.75,
        f.green_breadth <= 0.25,
    ])
    if enter_persistent:
        persistent_bear_active = True
        persistent_entry_session = t
        persistent_healthy_streak = 0
        audit("PERSISTENT_BEAR_ENTER", t, f, stress_days, stress_loss)

# Persistent Bear recovery
elif persistent_bear_active:
    healthy = (
        is_finite(f.r20) and f.r20 > 0
        and f.damaged_breadth <= 0.60
        and f.green_breadth >= 0.20
    )
    persistent_healthy_streak = persistent_healthy_streak + 1 if healthy else 0
    held = sessions_between(persistent_entry_session, t)
    if held >= 20 and persistent_healthy_streak >= 5:
        persistent_bear_active = False
        persistent_healthy_streak = 0
        ordinary_stress_start_session = t if ordinary_stress_active(t) else None
        audit("PERSISTENT_BEAR_EXIT", t, f, held)

# Resolve target allocation by priority
if systemic_emergency_active:
    target = 0.00
elif persistent_bear_active:
    target = 0.00
elif ordinary_stress_active(t):
    target = 0.40
else:
    target = 1.00

if target != previous_target_allocation:
    create_next_open_rebalance_orders(target)
    previous_target_allocation = target

persist_state_and_daily_snapshot(t)
```

8. Worked examples

8.1 2022 Persistent Bear entry

Observed feature on May 17, 2022	Value	Threshold	Pass?

Ordinary-stress duration	30 sessions	>= 30	Yes
Shadow return since stress start	-2.70%	<= -2%	Yes
Shadow r40	-5.37%	<= -3%	Yes
Positive-day share20	50%	<= 50%	Yes
Damaged breadth	78.3%	>= 75%	Yes
Green breadth	21.7%	<= 25%	Yes

Result: all predicates were true, so the live portfolio moved from 40% Wealth Core / 60% BIL to 0% Wealth Core / 100% BIL. The shadow remained fully invested and continued producing recovery evidence.

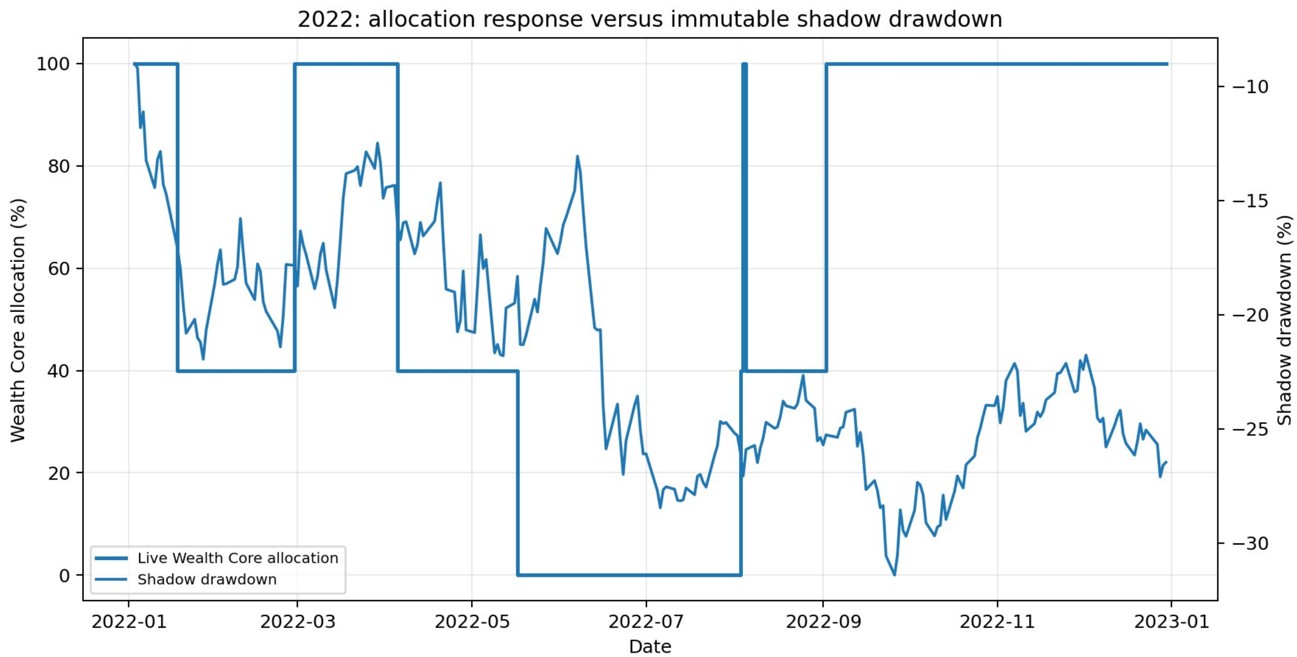

Figure 3. The 2022 Persistent Bear state removed the residual 40% equity allocation during the prolonged impaired period.

8.2 2022 exit

On August 3, 2022, the state had been active for 54 sessions. Shadow r20 was 3.41%, damaged breadth had improved to 33.3%, and green breadth had improved to 66.7%. These conditions had remained healthy for five consecutive sessions, satisfying recovery confirmation.

The controller did not infer recovery from the BIL-heavy live portfolio. It used the unchanged shadow portfolio, which is the only valid comparator for whether Wealth Core itself had recovered.

8.3 Fast systemic shock example

During a fast shock, the systemic classifier can move directly from NORMAL to 0% or override ORDINARY_STRESS. Example sequence: shadow drawdown crosses -10%, internal breadth collapses, five- or ten-session shadow loss exceeds the shock threshold, damaged breadth accelerates sharply, and SPY volatility/return confirms a market-wide panic. The controller stays fully defensive for at least 10 sessions and exits only after three healthy shadow sessions.

9. Research results and interpretation

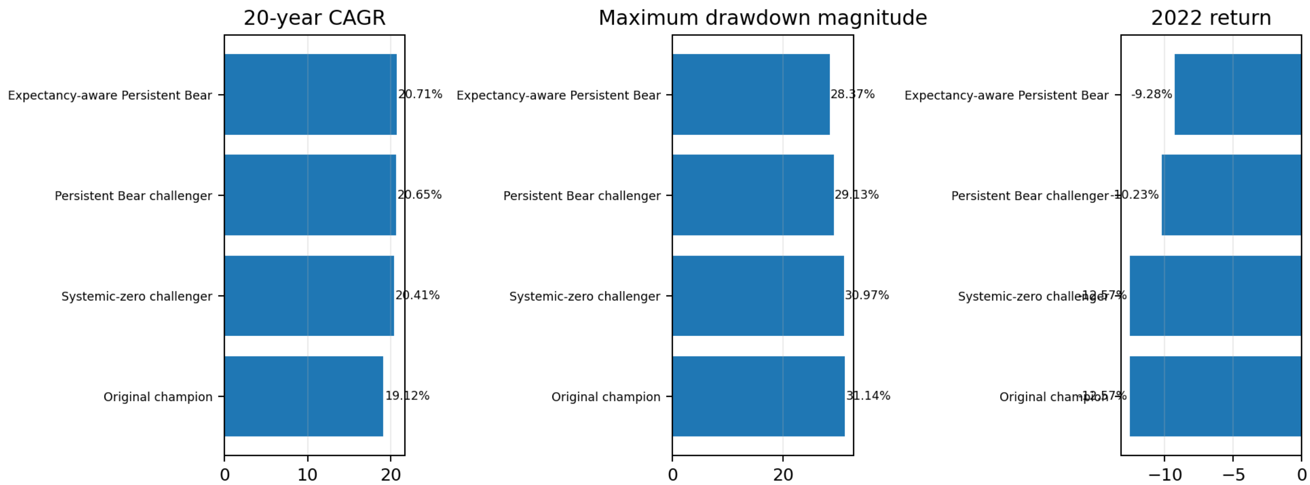

Figure 4. Research progression from the original champion to the latest expectancy-aware challenger.

Strategy	20-year CAGR	Max drawdown	2022 return	$1M ending wealth
Original champion	19.12%	-31.14%	-12.57%	$33.07M
Systemic-zero challenger	20.41%	-30.97%	-12.57%	$41.07M
Persistent Bear challenger	20.65%	-29.13%	-10.23%	$42.68M
Expectancy-aware Persistent Bear	20.71%	-28.37%	-9.28%	$43.14M

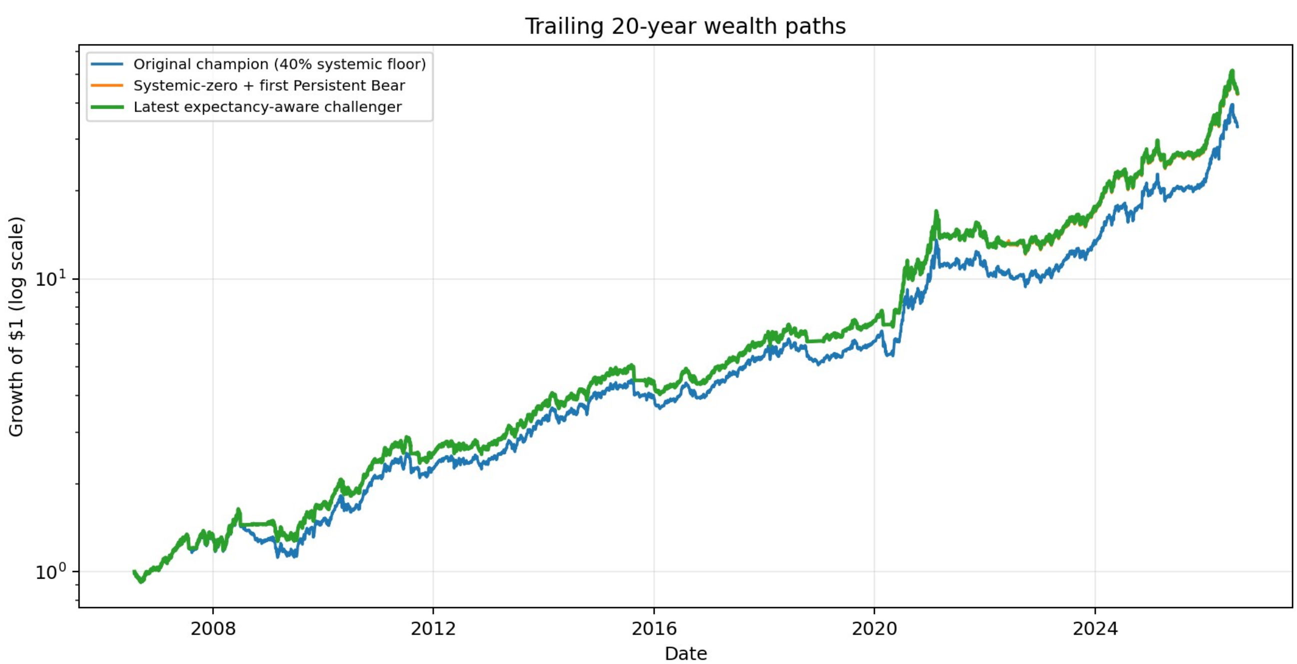

Figure 5. Cumulative wealth over the trailing 20-year research window, July 31, 2006 through July 31, 2026.

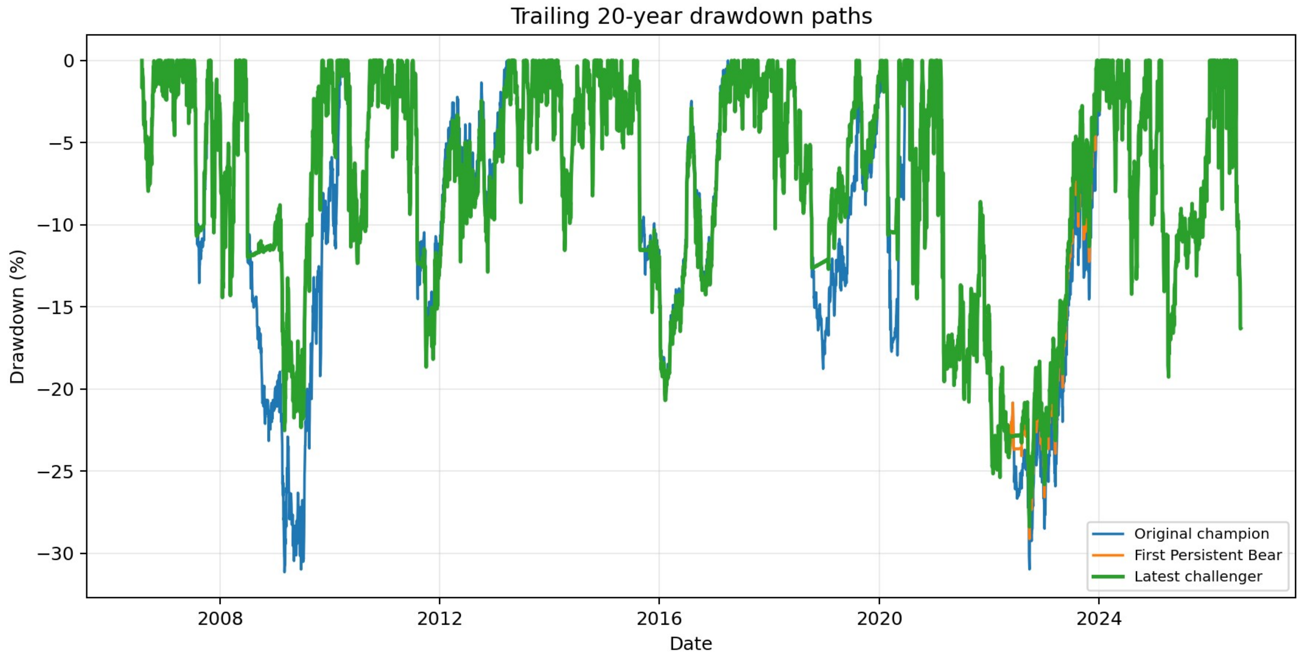

Figure 6. Drawdown comparison over the same trailing 20-year research window.

Interpretation. The latest challenger improved the research path by identifying four prolonged periods where the remaining 40% Wealth Core allocation continued to have negative expectancy. The additional gain is not evidence of certainty; it is a hypothesis supported by a small number of historical episodes and requires strict out-of-sample validation.

Entry	Exit	Context	Zero-state sessions
2001-09-06	2001-10-16	Post-dot-com deterioration	25
2008-10-02	2008-12-22	Global financial crisis	57
2018-12-07	2019-01-24	Q4 2018 selloff	32
2022-05-17	2022-08-03	2022 inflation/rate bear market	54

10. Testing and acceptance criteria

10.1 Unit tests

- Each feature formula at boundary values and with missing history.
- All entry predicates: one failing predicate must prevent entry.
- Stress duration off-by-one behavior.
- Consecutive recovery streak reset after one unhealthy session.
- Minimum-hold boundary: exit allowed only when held-session rule is met.
- State priority when systemic and Persistent Bear are simultaneously true.
- Stress reference reset on ordinary-stress exit and Persistent Bear recovery.
- Transition cost calculation for 100->40, 40->0, 0->40, and 0->100.
- No use of live NAV or live breadth in any signal calculation.

10.2 Golden-path tests

- Replay the frozen daily artifact and reproduce all eight Persistent Bear transition records exactly.

- Reproduce the selected trailing 20-year CAGR, maximum drawdown, 2022 return, and ending multiple within an agreed numeric tolerance.
- Verify that the shadow NAV is identical regardless of live controller state.
- Verify that restarting from a persisted checkpoint yields byte-equivalent subsequent transition logs.

10.3 Data-quality fail-safe behavior

Failure	Required behavior
Missing shadow NAV	Do not evaluate new transitions; raise critical alert.
Insufficient r40 history	Persistent Bear entry must evaluate false.
Invalid breadth denominator	Do not enter or recover; preserve current state and alert.
Missing defense price	Do not submit rebalance until executable quote exists; record blocked transition.
Duplicate session processing	Idempotently return prior result; never place duplicate orders.
State-store failure	Fail closed: do not change allocation without durable transition record.

10.4 Promotion gates

- Leave-one-episode-out validation.
- Rolling-origin or decade-by-decade parameter selection.
- One- and two-session signal-delay sensitivity.
- Next-open fills with spreads, slippage, dividends, and holidays.
- Parameter plateau analysis, not merely a single optimum.
- Parity across backtester, rehearsal, and live shadow.
- Prospective untouched shadow period.

11. Operational observability

Event / metric	Required fields
Daily snapshot	date, state, target allocation, actual allocation, live NAV, shadow NAV, all derived features, rule version
State entry	previous state, new state, all predicates and values, stress reference, decision timestamp
State exit	held sessions, healthy streak, recovery feature values, destination allocation
Rebalance order	target allocation, estimated trade value, instrument list, decision session, intended fill session
Fill	instrument, quantity, price, timestamp, commission, spread/slippage estimate
Parity metric	difference between live shadow and certified shadow target/NAV
Alert	missing data, stale data, blocked order, state-store error, parity breach

Recommended API fields for a progress endpoint: controller_state, target_wealth_core_allocation, actual_wealth_core_allocation, shadow_drawdown, stress_duration, shadow_r20, shadow_r40, positive_day_share20, damaged_breadth, green_breadth, persistent_healthy_streak, systemic_healthy_streak, last_transition, and configuration_hash.

12. Known limitations and next research steps

- The selected rule activated only four historical Persistent Bear episodes.
- Only 12 of 2,592 tested parameter configurations improved all chosen targets; the viable plateau is not yet broad.

- The frozen research path is an overlay, not a full certified replay through the latest production engine.
- Results depend on the exact definition of damaged and green holdings.
- Tax consequences, market impact, and real-world BIL execution were not fully modeled.
- The current rule uses fixed thresholds and does not estimate uncertainty or changing structural regimes.
- Recovery may still be vulnerable to short-lived bear-market rallies.

Recommended next experiment. Keep this specification frozen as Challenger v2. Perform leave-one-episode-out and rolling-origin validation before changing thresholds further. Any future iteration should be compared against this exact version and should explain which historical decisions changed and why.

13. Engineering implementation checklist

☐ Version and freeze the Wealth Core shadow implementation and breadth definitions.

☐ Implement feature calculations with session-aware rolling windows.

☐ Persist controller state and stress-reference indices transactionally.

☐ Implement ordinary, systemic, and Persistent Bear classifiers as independent pure functions.

☐ Implement priority-based allocation arbitration.

☐ Generate next-open rebalance orders with idempotency keys.

☐ Model the defensive sleeve as a total-return instrument.

☐ Record transition costs on both traded sleeves.

☐ Expose full telemetry and machine-readable reason codes.

☐ Build unit, golden-path, restart, and parity tests.

☐ Reproduce the frozen transition table and performance metrics.

☐ Run promotion-gate validation before enabling production capital.

Appendix A. Configuration object

```
{
  "strategy_id": "wealth_core_expectancy_persistent_bear_v2",
  "ordinary": {
    "trigger_drawdown": -0.155,
    "target_allocation": 0.40
  },
  "systemic": {
    "internal_drawdown": -0.10,
    "damaged_min": 0.85,
    "green_max": 0.20,
    "r5_max": -0.05,
    "r10_max": -0.08,
    "damaged_delta5_min": 0.40,
    "vol_acceleration_min": 0.04,
    "spy_r20_max": -0.01,
    "fallback_shadow_r10": -0.10,
    "target_allocation": 0.00,
    "min_hold_sessions": 10,
    "recovery_confirmation": 3
  },
  "persistent_bear": {
```

```
    "min_stress_sessions": 30,
    "loss_since_stress_max": -0.02,
    "shadow_r40_max": -0.03,
    "positive_day_share20_max": 0.50,
    "damaged_min": 0.75,
    "green_max": 0.25,
    "target_allocation": 0.00,
    "min_hold_sessions": 20,
    "recovery_confirmation": 5,
    "recovery_shadow_r20_min_exclusive": 0.00,
    "recovery_damaged_max": 0.60,
    "recovery_green_min": 0.20
  },
  "execution": {
    "decision_timing": "after_close",
    "fill_timing": "next_open",
    "cost_per_side": 0.001,
    "defensive_asset": "BIL"
  }
}
```


Appendix B. Frozen research transitions

Date	Event	Duration	Ref return	Shadow DD	r20	r40	Pos20	Damaged	Green
2001-09-06	enter	52	-6.00%	-20.82%	-3.97%	-5.79%	45%	76.2%	14.3%
2001-10-16	exit	25	2.41%	-18.92%	7.84%	-2.97%	55%	26.3%	63.2%
2008-10-02	enter	51	-14.56%	-27.93%	-6.72%	-11.98%	45%	75.0%	6.2%
2008-12-22	exit	57	-8.39%	-33.97%	5.35%	1.54%	65%	26.3%	68.4%
2018-12-07	enter	31	-5.14%	-21.51%	-5.93%	-10.33%	40%	76.2%	23.8%
2019-01-24	exit	32	-0.65%	-22.02%	8.00%	-0.15%	70%	45.8%	50.0%
2022-05-17	enter	30	-2.70%	-18.32%	-3.91%	-5.37%	50%	78.3%	21.7%
2022-08-03	exit	54	-9.43%	-26.02%	3.41%	-13.40%	50%	33.3%	66.7%

Appendix C. Source artifacts

- persistent_bear_expectancy_challenger_daily.csv
- persistent_bear_expectancy_transitions.csv
- persistent_bear_expectancy_metrics.csv
- persistent_bear_expectancy_grid.csv
- persistent_bear_iteration2_summary.csv
- best_systemic_confirmation_daily.csv and associated systemic-forensics report
- defensive_composite.csv and progressive-controller research report

End of specification.